PRACTICAL

JAVA OOP PROGRAMS

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Subject	Java Programming / Object-Oriented Programming
Programs	Student, Employee, BankAccount, Rectangle, and Book

This practical record contains the Java source code and corresponding sample console output for all five programs.

1. Student Class

Objective: Create three Student objects and display each student's ID, name, and age.

Concepts: Class, objects, instance variables, and methods.

Java Source Code

```
class Student {
    int studentId;
    String name;
    int age;

    void displayDetails() {
        System.out.println("Student ID: " + studentId);
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
        System.out.println("--------------------");
    }

    public static void main(String[] args) {

        Student s1 = new Student();
        s1.studentId = 101;
        s1.name = "Arun";
        s1.age = 19;

        Student s2 = new Student();
        s2.studentId = 102;
        s2.name = "Sai";
        s2.age = 20;

        Student s3 = new Student();
        s3.studentId = 103;
        s3.name = "Yuvan";
        s3.age = 19;

        s1.displayDetails();
        s2.displayDetails();
        s3.displayDetails();
    }
}
```

Sample Output

```
Student ID: 101
Name: Arun
Age: 19
--------------------
Student ID: 102
Name: Sai
Age: 20
--------------------
Student ID: 103
Name: Yuvan
Age: 19
--------------------
```

2. Employee Class

Objective: Initialize employee ID, name, and salary using a parameterized constructor, then display the details.

Concepts: Parameterized constructor and object initialization.

Java Source Code

```
class Employee {
    int employeeId;
    String name;
    double salary;

    Employee(int id, String n, double s) {
        employeeId = id;
        name = n;
        salary = s;
    }

    void display() {
        System.out.println("Employee ID: " + employeeId);
        System.out.println("Name: " + name);
        System.out.println("Salary: " + salary);
    }

    public static void main(String[] args) {

        Employee e1 = new Employee(101, "Arun", 30000);
        Employee e2 = new Employee(102, "Sai", 35000);

        e1.display();

        System.out.println();

        e2.display();
    }
}
```

Sample Output

```
Employee ID: 101
Name: Arun
Salary: 30000.0

Employee ID: 102
Name: Sai
Salary: 35000.0
```

3. BankAccount Class

Objective: Create a bank account and demonstrate depositing money, withdrawing money, and displaying the balance.

Concepts: Constructors, methods, and conditional statements.

Java Source Code

```
class BankAccount {
    int accountNumber;
    double balance;

    BankAccount(int acc, double bal) {
        accountNumber = acc;
        balance = bal;
    }

    void deposit(double amount) {
        if (amount > 0) {
            balance = balance + amount;
            System.out.println("Deposited: " + amount);
        } else {
            System.out.println("Invalid deposit amount");
        }
    }

    void withdraw(double amount) {
        if (amount > 0 && amount <= balance) {
            balance = balance - amount;
            System.out.println("Withdrawn: " + amount);
        } else {
            System.out.println("Insufficient balance or invalid amount");
        }
    }

    void displayBalance() {
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Balance: " + balance);
    }

    public static void main(String[] args) {

        BankAccount b1 = new BankAccount(12345, 5000);

        b1.displayBalance();

        b1.deposit(2000);
        b1.withdraw(1500);

        System.out.println("Final Details:");
        b1.displayBalance();
    }
}
```

Sample Output

```
Account Number: 12345
Balance: 5000.0
Deposited: 2000.0
Withdrawn: 1500.0
Final Details:
Account Number: 12345
Balance: 5500.0
```

4. Rectangle Class

Objective: Calculate and display the area and perimeter of a rectangle.

Concepts: Parameterized constructor, methods, and arithmetic calculations.

Java Source Code

```
class Rectangle {
    double length;
    double width;

    Rectangle(double l, double w) {
        length = l;
        width = w;
    }

    void calculate() {
        double area = length * width;
        double perimeter = 2 * (length + width);

        System.out.println("Length: " + length);
        System.out.println("Width: " + width);
        System.out.println("Area: " + area);
        System.out.println("Perimeter: " + perimeter);
    }

    public static void main(String[] args) {

        Rectangle r1 = new Rectangle(10, 5);

        r1.calculate();
    }
}
```

Sample Output

```
Length: 10.0
Width: 5.0
Area: 50.0
Perimeter: 30.0
```

5. Book Class

Objective: Initialize a book's ISBN, title, and price using a parameterized constructor, then display the details.

Concepts: Parameterized constructor and object creation.

Java Source Code

```
class Book {
    String ISBN;
    String title;
    double price;

    Book(String i, String t, double p) {
        ISBN = i;
        title = t;
        price = p;
    }

    void display() {
        System.out.println("ISBN: " + ISBN);
        System.out.println("Title: " + title);
        System.out.println("Price: " + price);
    }

    public static void main(String[] args) {

        Book b1 = new Book("9781234567890",
                           "Java Programming", 450);

        b1.display();
    }
}
```

Sample Output

```
ISBN: 9781234567890
Title: Java Programming
Price: 450.0
```